

ELL715 - Assignment 4

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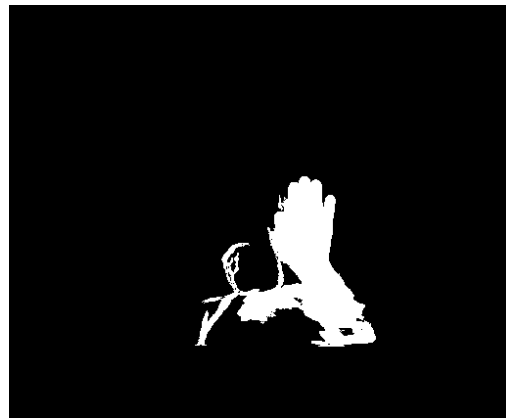
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1 Part1: Paper Implementation

1.1 Implementation of Silhouette Tunnel



(a) original-Compass

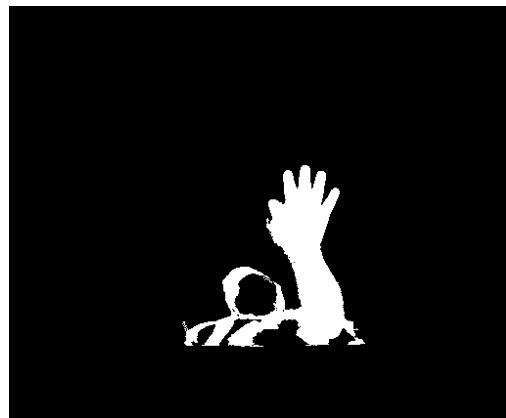


(b) silhouette mask - Compass

Figure 1: Extraction of silhouette mask from compass



(a) original-Piano



(b) silhouette mask - Piano

Figure 2: Extraction of silhouette mask from piano



(a) original-Push

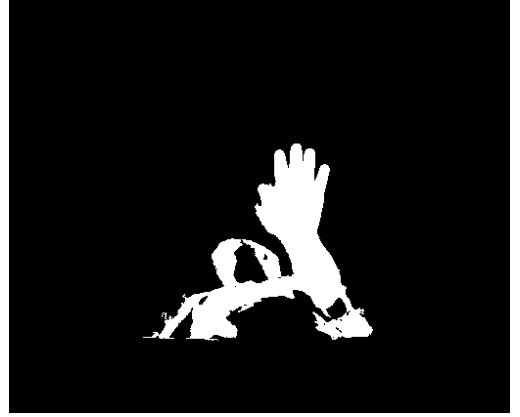


(b) silhouette mask - Push

Figure 3: Extraction of silhouette mask from Push



(a) original-UCDO



(b) silhouette mask - UCDO

Figure 4: Extraction of silhouette mask from UCDO

Each pixel within the silhouette tunnel is represented by a 14-dimensional feature vector:

$$\mathbf{f} = [x, y, t, z, d_E, d_W, d_N, d_S, d_{NE}, d_{NW}, d_{SE}, d_{SW}, dT^+, dT^-]^T$$

where:

- x, y, t denote the spatial and temporal coordinates of the pixel,
- z is the depth value,
- $d_E, d_W, d_N, d_S, d_{NE}, d_{NW}, d_{SE}, d_{SW}$ represent directional distances from the pixel to the nearest background boundary in the eight cardinal directions, and
- dT^+, dT^- denote the forward and backward temporal persistence of the silhouette pixel across time.

Directional distances and temporal runs are efficiently computed using Numba-accelerated kernels for parallel processing.

All feature vectors from the sequence are collected into a feature matrix:

$$\mathbf{F} = [\mathbf{f}_1 \quad \mathbf{f}_2 \quad \dots \quad \mathbf{f}_N] \in \mathbb{R}^{14 \times N}$$

and are normalized feature-wise to the range $[0, 1]$ as:

$$\mathbf{F}_{\text{norm}} = \frac{\mathbf{F} - \mathbf{F}_{\min}}{\mathbf{F}_{\max} - \mathbf{F}_{\min} + \epsilon}$$

where ϵ is a small constant (e.g., 10^{-6}) to avoid division by zero.

1.2 Extraction of Covariance Matrix

The silhouette tunnel provides us with a rich representation of the pixel features over time. To analyze the variability and relationships between these features, we compute the covariance matrix of the normalized feature matrix:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{F}_{\text{norm}} \mathbf{F}_{\text{norm}}^T$$

where $\mathbf{C} \in \mathbb{R}^{14 \times 14}$ is the covariance matrix capturing the pairwise feature correlations across all pixels in the silhouette tunnel across all the frames.

Note: In the paper, the denominator of the above equation is given as N . However, to obtain an unbiased estimate of the covariance, we use $N-1$ in our implementation.

We then just take the upper triangular part of the covariance matrix (including the diagonal) and vectorize it to form our final feature descriptor for the video sequence:

$$\mathbf{d} = \text{vec}(\mathbf{C}_{\text{upper}})$$

where $\mathbf{d} \in \mathbb{R}^{105}$ is the final feature descriptor capturing the relationships between the pixel features over time.

1.3 Temporal Hierarchical Covariance Descriptor

Hierarchical Temporal Segmentation: The sequence is divided into multiple temporal segments at different levels of granularity. For a hierarchy depth of $L = 3$, the division results in:

$$1 + 2 + 4 = 7 \text{ temporal partitions.}$$

Each level $\ell \in \{1, 2, \dots, L\}$ contains $2^{\ell-1}$ equal-length temporal segments.

For a given level ℓ and segment s , the subset of feature vectors belonging to that temporal range is denoted:

$$F_{\ell,s} = \{f_n \mid t_n \in [s/S_\ell, (s+1)/S_\ell)\}, \quad \text{where } S_\ell = 2^{\ell-1}$$

Covariance Representation per Segment: For each temporal segment $F_{\ell,s}$, we compute its empirical covariance matrix:

$$C_{\ell,s} = \frac{1}{N_{\ell,s} - 1} \sum_{n=1}^{N_{\ell,s}} (f_n - \mu_{\ell,s})(f_n - \mu_{\ell,s})^\top$$

where

$$\mu_{\ell,s} = \frac{1}{N_{\ell,s}} \sum_{n=1}^{N_{\ell,s}} f_n$$

is the mean feature vector within that segment, and $N_{\ell,s}$ is the number of samples in that segment.

Compact Covariance Encoding: Since $C_{\ell,s} \in \mathbb{R}^{D \times D}$ is symmetric, only its upper-triangular elements are retained to avoid redundancy. The unique elements are vectorized as:

$$C_{\ell,s}^{ut} = \text{vec}_{ut}(C_{\ell,s}) \in \mathbb{R}^{\frac{D(D+1)}{2}}$$

where $\text{vec}_{ut}(\cdot)$ denotes concatenation of the upper-triangular entries.

Final Descriptor: All such upper-triangular covariance vectors are concatenated across the hierarchy to form the final temporal hierarchical descriptor:

$$\text{Descriptor}_{TH} = [C_{1,1}^{ut}, C_{2,1}^{ut}, C_{2,2}^{ut}, \dots, C_{L,2^L-1}^{ut}] \in \mathbb{R}^{(2^L-1) \cdot \frac{D(D+1)}{2}}$$

For $D = 14$ and $L = 3$, this yields:

$$\text{Descriptor}_{TH} \in \mathbb{R}^{7 \times 105} = \mathbb{R}^{735}$$

Implementation Note: Numba’s `jit(parallel=True)` acceleration is used to parallelize computations, yielding a 5–10× runtime speedup.

References